University of St.Gallen

3,130/4,130 CORPORATE FINANCE

MOCK EXAM

PROF. DR. MARKUS SCHMID

Please note:

- You can reach a maximum of **90 points**. One point equals about one minute.
- **Only one** of the possible answers in each problem set is correct. Please tick the respective box on the **answer sheet**. Answers recorded in the compilation of problems **will not be graded**.
- Only the boxes ticked on the answer sheet will be graded, additional comments or calculations will be disregarded.
- Ticking the correct box, by marking it with an "**x**" will result in the full points awarded for this (sub-) problem. Not ticking any box, ticking a wrong box, or ticking several boxes will yield **zero points** (there are **no** minus points).
- You are expected to choose the **most likely** result if you have to do any calculations. If not stated otherwise in the question, round your **final results** to **two decimal places.** If **necessary**, round your intermediate results to **four decimal places**.
- A **table** containing the **probability values for the standard normal distribution** as well as a compilation of formulas are given on the last three pages of this compilation of problems.

Good luck!

1. True-or-false problems 1 point per sub-problem (20 points in total)

Is the statement true or false?

Problem 1.1
Monthly equity returns are close to a log-normal distribution while long-term returns are approximately normally distributed.

a) true
b) false

Problem 1.2
Risk shifting, debt overhang, and playing for time are potential consequences of conflicts of interest between debt and equity.

a) true
b) false

Problem 1.3
Value destroying (diversifying) acquisitions are a possible outcome of poor corporate governance.

a) true
b) false

Problem 1.4
One reason for hedging is to avoid losses which cannot be completely offset against gains. Therefore, firms may have to pay higher (expected) taxes if they do not hedge.

a) true
b) false

Problem 1.5

When estimating the weighted-average cost of capital, debt and equity can enter at either the nominal or the market value, as long as a choice is made among these two options and kept consistent over the entire calculation.

a) true
b) false

Problem 1.6

"Augmented dividend" refers to the dividend payments adjusted for stock repurchases. This is an important adjustment to just using cash dividends in a dividend discount model because the model may otherwise estimate a false equity value.

a) true
b) false

Problem 1.7

The buyer of a future, i.e., holding a short position, makes a gain if the price of the underlying exceeds the future price at time T.

a) true
b) false

Problem 1.8

In order to calculate the entity value of a firm, free cash flows are discounted by the cost of equity in the DCF method.

a) true
b) false

Problem 1.9

The expected value of the log of the value of an investment at the end of the holding period is equal to the log of the initial investment plus the sum of all the expected continuous returns.

a) true
b) false

Problem 1.10

Net stock issues by U.S. nonfinancial corporations between 1991 and 2017 were negative in most years, contributing to an increase in leverage ratios.

a) true
b) false

Problem 1.11

The disposition effect states that investors are generally risk-averse regarding gains but risk-seeking regarding losses. As a consequence, they hold on to “loser stocks” for too long, while selling “winner stocks” too early.

a) true
b) false

Problem 1.12

The variance of a random variable is equal to the probability weighted sum of the squared deviations between the realizations of the random variable and its expected value.

a) true
b) false

Problem 1.13

Empirical evidence suggests that investors demand significantly higher expected returns on stocks excluded by environmental screens compared to firms without such environmental concerns.

a) true
b) false

Problem 1.14

The Security Market Line (SML) is the graphical illustration of the Capital Asset Pricing Model (CAPM).

a) true
b) false

Problem 1.15

Financial markets reward a capital commitment over time, because people are impatient. Financial markets reward risk, because people are risk averse and thus prefer a secure position over a risky one.

a) true
b) false

Problem 1.16

The difference between the net present value (NPV) and the present value (PV) is that one uses the nominal interest rate in order to calculate the NPV whereas the latter one uses the real interest rate in order to calculate to the PV.

a) true
b) false

Problem 1.17

If firm value in case of liquidation is higher than in case of going concern, debt holders would delay the liquidation of the firm in order to gamble for resurrection.

a) true
b) false

Problem 1.18

The Adjusted Present Value Method (APV) is most useful when side-costs are numerous and important.

a) true
b) false

Problem 1.19

Venture Capital companies generally provide financing sufficient to cover all development expenses in one stage.

a) true
b) false

Problem 1.20

Firms try to avoid taxes. In order to address asymmetric tax treatment, firms with higher growth opportunities use more derivatives to hedge so that they can increase their costs of financial distress which are tax deductible.

a) true
b) false

Problem 2 **3 points**

Mister Smith expects periodic payments which will be paid in perpetuity. The payments grow at a rate of 2% p.a., are paid once a year, and occur at the end of each year. The first payment in the amount of EUR 300 occurs in exactly one year. Assume that the appropriate discount rate is 4.5% p.a.

What is the most likely present value of these payments?

a) EUR 6,666
b) EUR 12,000
c) EUR 12,240
d) EUR 16,666
e) EUR 18,000

$$PV = \frac{300}{0.045 - 0.02} = \underline{\underline{12'000}}$$

Problem 3 **3 points**

Ken Duck has just won the Swiss lottery, paying CHF 500,000 per year for 2 years. He is to receive his first payment a year from now. The state advertises this as the Million Swiss Lottery because CHF 500,000 x 2 = CHF 1,000,000.

If the interest rate is 8%, what is the most likely true value of the lottery? Round your result to the nearest integer.

a) CHF 428,669
b) CHF 462,963
c) CHF 857,339
d) CHF 891,632
e) CHF 926,438

$$PV = \frac{500'000}{1.08} + \frac{500'000}{1.08^2} = 462'962.963 + 428'669.4102 = \underline{891'632.37}$$

Problem 4 **4 points**

Which statement regarding the case study of Cox Communications, Inc. is correct?

a) Clement is reluctant to issue equity for the only reason that this would decrease the family's economic ownership of Cox.
b) If Cox Communications, Inc. decides to issue debt rather than equity to fund the acquisition of Gannett, the direct and indirect costs of a debt issuance will be more than that for issuing equity.
c) The management of Cox Communications, Inc. was concerned about increasing financial leverage. This financial instrument would act against the goal of the Cox family to maintain a high debt rating.
d) Because internal cash flows of Cox Communications, Inc. are sufficient to fund all acquisitions already announced, it would be rational for the management to temporally increase the debt ratio even though this would increase the threat of a debt downgrade.
e) None of the answers a) to d) are correct.

Problem 5 4 points

Mr. Welch, CEO of the Frost company, has CHF 60'000 in cash to invest and he is considering two mutually exclusive investment projects. The first project requires an immediate cash-outflow of CHF 50'000 in t=0 and pays back CHF 22'000 in t=1 and CHF 38'000 in t=2. The second project requires a cash-outflow of CHF 6'000 in t=0 and pays back CHF 8'000 in t=2. The first project is considered to be rather risky and alternative investments with the same risk earn a return of 10% in the capital market. The cash-flows of the second project are expected to be less risky and alternative investments with the same risk earn a return of 4% in the capital market.

Based on an NPV analysis, how should Mr. Welch decide?

Mr. Welch should invest in …

a) Project 1
b) Project 2
c) Project 1 and 2
d) None of the projects.
e) None of the answers a) to d) are correct.

$$NPV_1 = -50000 + \frac{22000}{1.1} + \frac{38000}{1.1^2} = 1404.96$$

$$NPV_2 = -6000 + \frac{8000}{1.04^2} = 1396.45$$

Even though Mr. Welch disposes of sufficient funding to invest into both projects, and both have a positive NPV, you will only invest into Project 1 with the highest NPV as the projects are mutually exclusive.

Problem 6 **3 points**

Which statement is **correct**?

a) In the Gordon Growth Model, the value of a company is calculated by dividing today's dividend by the difference between the growth rate and the discount rate.
b) The Gordon Growth Model can easily accommodate changing dividend growth rates and also works for non-dividend paying stocks.
c) The rate of organic growth of a firm is defined as the growth rate which is achieved by increasing equity capital.
d) If both the payout ratio and the growth rate remain the same, a reduction in risk will lead to a higher value of a given firm.
e) None of the answers a) to d) are correct.

Problem 7 **4 points**

Shares in a large building materials stock, Jame Hardie Corporation (JHC), were trading for CHF 5.81 on 15 January 2020. JHC had paid a total dividend of around 32 cents per share in the previous year. Assume that the required return on Boral Inc. is 8.00% and appropriate for valuing JHC.

Based on the Gordon Growth Model, what perpetual future dividend growth rate is most likely being used by the market to price JHC?

a) 2.36%
b) 8.67%
c) 11.41%
d) 15.80%
e) 45.25%

$$5.81 = \frac{0.32 \cdot (1 + g)}{0.08 - g} \rightarrow g = 0.0236$$

Problem 8 **5 points**

Calculate the correlation coefficient between the stock of Ehn company and the market.

risk-free interest rate:	3%
standard deviation of σ_{Ehn}:	12%
standard deviation σ_{Market}:	8%
expected return r_{Ehn}:	8%
market risk premium:	6%
estimated growth rate:	4%
payout ratio:	40%

The most likely correlation coefficient is …

a) 0.50
b) 0.56
c) 0.88
d) 1.11
e) 1.25

$$CAPM: r = r_f + \beta \cdot (r_M - r_f)$$

$$\beta = \frac{0.08 - 0.03}{0.06} = 0.8333$$

$$\rho = \frac{0.8333 \cdot 0.08}{0.12} = 0.556$$

Problem 9 **6 points**

Consider shares in two companies, X and Y, with the following characteristics:

	$E(R_i)$	σ_i
Share X	10%	20%
Share Y	30%	50%

The correlation between X and Y (i.e., ρ_{XY}) = 0.4

9.1 What is the most likely expected return of a portfolio that invests 25% in share X and 75% in share Y? (**3 points)**

a) 15.00%
b) 19.50%
c) 22.50%
d) 25.00%
e) 42.50%

$$E(r) = 0.25 \cdot 0.1 + 0.75 \cdot 0.3 = 0.25$$

9.2 What is the most likely standard deviation of a portfolio that invests 25% in share X and 75% in share Y? **(3 points)**

a) 28.10%
b) 31.64%
c) 40.00%
d) 54.61%
e) 63.72%

$${\sigma_{E(R)}}^2 = 0.25^2 \cdot 0.2^2 + 0.75^2 \cdot 0.5^2 + 2 \cdot 0.25 \cdot 0.75 \cdot (0.4 \ \cdot 0.2 \cdot 0.5) = 0.1581$$

$$\sigma_{E(R)} = \sqrt{0.1581} = 0.40$$

Problem 10 **4 points**

The Carpenter Corporation will generate an EBIT of 300m CHF each year (until infinity). The cost of capital of the unlevered firm is 8% p.a. and the corporate tax rate is 35%. The present value of all tax advantages (tax shield) is 120m CHF. Calculate the entity value.

The most likely entity value of the Carpenter Corporation is …

a) CHF 1’090.00 m
b) CHF 1'432.50 m
c) CHF 2‘557.50 m
d) CHF 3’870.00 m
e) CHF 4’230.00 m

$$w(Entity) = \frac{300m * (1-0.35)}{0.08} + 120m = 2557.50mCHF$$

Problem 11 **4 points**

Looking at the case study "Valuing Snap After the IPO Quiet Period", which statement regarding the valuation of Snap using Morgan Stanley's (original) forecasts is correct?

a) Using a long forecast period of 10 years prevents Snap's revenue growth rate from a dramatic decline.
b) Because the cash flows must be in a steady state before calculating a terminal value, analysts use a long forecast period to accommodate very low growth rates in the early years and allow sufficient time for growth to increase in later years.
c) Snap does not pay taxes until 2024; this recent change in the amount of taxes paid is the unique reason why the cash flows in the terminal year may not be in a steady state.
d) Snap is "asset light" i.e., capital expenditures and working capital are modest compared to its revenues forecast, as Snap uses third-party infrastructure partners like Google Cloud to host its services and to avoid significant capital expenditures to support revenue generating activities.
e) None of the answers a) to d) are correct.

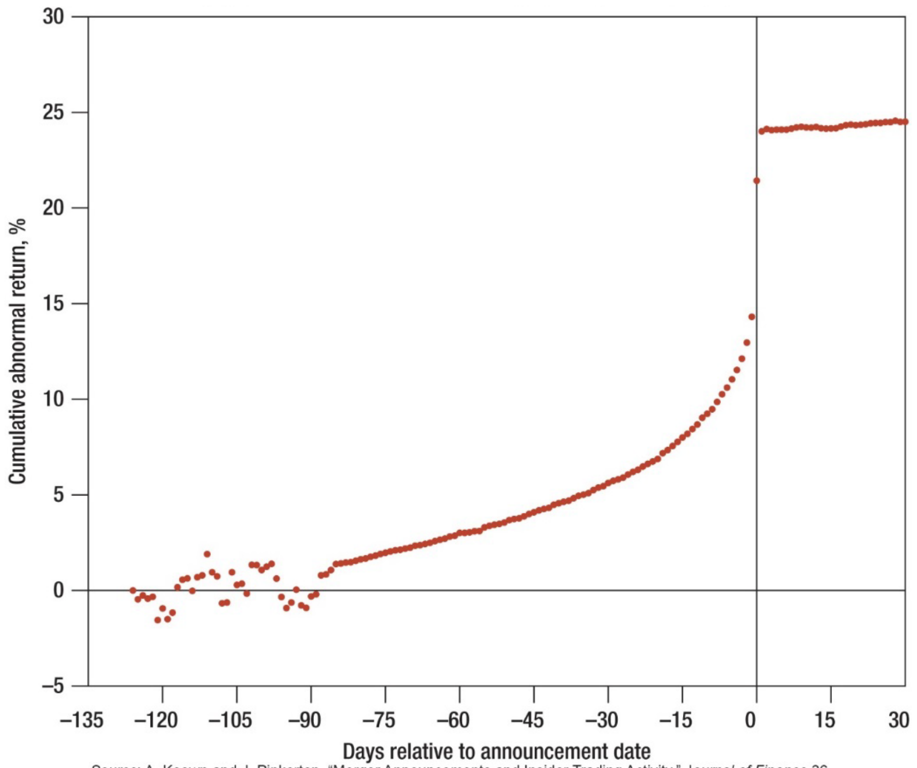
Problem 12 **4 points**

Which of the following are potentially effective corporate governance mechanisms:

a) A pay structure which aligns top managers' incentives with the interests of shareholders; an independent board of directors; a strong anti-takeover protection.
b) An independent board of directors; a combined CEO and chairman position; a powerful CEO sitting on the nominating and compensation committees.
c) A pay structure which aligns top managers' incentives with the interests of shareholders; an active takeover market; an independent board of directors; product market competition.
d) A pay structure which aligns top managers' incentives with the interests of shareholders; an independent board of directors; high research and development expenditures, capital expenditures, and acquisition spendings.
e) Neither of the above.

Problem 13 **4 points**

Consider the following chart which shows the price reaction to takeover announcements of around 17'000 takeovers which are publicly announced at day 0 (where a vertical line intersects the horizontal axis).



Source: A. Keown and J. Pinkerton, "Merger Announcements and Insider Trading Activity," *Journal of Finance* 36 (September 1981), pp. 855–869. © 1981. Used with permission of John Wiley and Sons, via Copyright Clearance Center. Updates courtesy of Jinghua Yan

Which of the following statements is correct?

a) In the period 120 to 90 days prior to the takeover announcement, the overall expectations of market participants regarding the average takeover were pessimistic.

b) Although banks and regulators monitor trading in stocks of companies involved in takeovers, information leakage and insider trading might still contribute to this pattern.

c) Due to the fact that market prices adjust immediately after the announcement, it can be inferred from the chart that the market was strong-form efficient.

d) The pattern above is evidence against the possible existence of quant traders attempting to predict takeover attempts.

e) None of the answers a) to d) are correct.

Problem 14 **4 points**

Assume an APT model with three factors. Stock A has an expected return of 16%. The beta of factor 1 is 0.9, the beta of factor 2 is 0.6, and the beta of factor 3 is 1.1. The risk premium of factor 1 is 5%, the risk premium of factor 2 is 2% and the interest rate is 6%.

What should be the most likely risk premium of factor 3 if there is no arbitrage?

a) 2.84%
b) 2.99%
c) 3.20%
d) 3.41%
e) 3.91%

$$r_A = i + \beta_1 \cdot r_1 + \beta_2 \cdot r_2 + \beta_3 \cdot r_3$$
$$0.16 = 0.06 + 0.9 \cdot 0.05 + 0.6 \cdot 0.02 + 1.1 \cdot r_3$$
$$1.1 \cdot r_3 = 0.16 - 0.06 - 0.045 - 0.012$$
$$1.1 \cdot r_3 = 0.043$$
$$r_3 = 0.03909 = 0.0391 = 3.91\%$$

Problem 15 **5 points**

The Fruit Company is considering a project that will result in initial after tax cash savings of CHF 5 million at the end of the first year. These savings will grow at a rate of 5 percent per year. The firm has a debt-equity ratio of 0.5, a cost of equity of 29.2 percent, and a cost of debt of 10 percent. The cost-saving proposal is closely related to the firm's core business, so it can be considered to have the same risk as the firm. Assume a 34 percent tax rate and initial project costs are CHF 29 million.

Should the firm take on the project?

a) Yes, the NPV is positive.
b) Yes, the PV is zero.
c) No, the NPV is negative.
d) Indifferent, the initial costs equal the present value.
e) None of the answers a) to d) are correct.

WACC = (E/V) *rE + (D/V)*rD*(1-t) = 2/3 *0.292 + 1/3*0.1*(1-0.34) = <u>21.67%</u>

$$PV = \frac{5 million}{0.2167 - 0.05}$$ ≈ 30 Million -> NPV = 30 -29 = <u>1 million</u>

Problem 16 **4 points**

Assume that we obtained the following results from estimating a two-factor model:

$$CF = 25 + 3F_{Curr} - 6F_{int}$$

CF: Cash flow
F_{Curr}: percentage change in the EUR/USD exchange rate over the next year
F_{int}: percentage change in three-month LIBOR from now until one year from now

The expected variance / covariance structure of the factor exposures is:

$$\sigma_{curr}{}^2 = 0.021 \quad \sigma_{int}{}^2 = 0.032 \qquad cov_{curr/int} = 0.006$$

What is the most likely factor-based volatility of the firm's cash flows?

a) 0.01
b) 0.17
c) 1.06
d) 1.25
e) 1.60

$$\sigma_{CF} = \sqrt{3^2 \cdot 0.021 + 6^2 \cdot 0.032 + 2 \cdot 3 \cdot (-6) \cdot 0.006} = 1.06$$

Problem 17 **4 points**

Marvin Enterprises shows the following information in its prospectus for its IPO. Marvin has 900,000 shares of common stocks outstanding. Before this offering, there has been no public market for the common stock.

	Price to public	Underwriting discount
Per share	\$80.00	\$5.60
Total	\$72,000,000	\$5,040,000

The proceeds to company are computed before deducting administrative fees payable by the company estimated at \$820,000. One day-post IPO, Marvin shares traded at \$105.

What was the total dollar cost of Marvin issue?

a) USD 27,904,445
b) USD 27,995,555
c) USD 28,360,000
d) USD 33,035,555
e) USD 33,400,000

$$Underwriting\ fee = \$5,040,000$$
$$Administrative\ costs = \$820,000$$
$$Underpricing\ cost = 900,000 \cdot (105 - 80) = \$22,500,000$$
$$Total\ costs = \$28,360,000$$

Problem 18 5 points

The table reports results from OLS regressions of the natural logarithm of Tobin's Q on a dummy variable whether a company has a staggered board (i.e., a board in which each director is elected for three years and thus one third of the board of directors is up for election in any given year) and different sets of control variables. The sample includes up to 2,961 listed US firms over the time period 1991-2013. The table reports coefficient estimates and t-statistics (in parentheses). *, **, *** indicate statistical significance at the 10%, 5%, 1% level, respectively.

	(1)		(2)		(3)		(4)	
Staggered Board	-0.023	**	-0.025	**	-0.015		-0.015	
	(-2.110)		(-2.292)		(-1.457)		(-1.302)	
Log (Total Assets)	-0.021	***	-0.018	***	-0.085	***	-0.106	***
	(-4.697)		(-3.655)		(-9.595)		(-10.197)	
Log (Age)	-0.012		-0.010		-0.021	***	-0.017	**
	(-1.490)		(-1.117)		(-2.760)		(-2.017)	
Delaware	0.023	*	0.022	*	0.027	**	0.029	**
	(1.908)		(1.797)		(2.278)		(2.236)	
Return on Assets	1.899	***	2.044	***	1.602	***	1.689	***
	(20.674)		(20.049)		(15.085)		(13.065)	
Capital Expenditures	-0.131		-0.267	**	0.014		-0.239	**
	(-1.280)		(-2.287)		(0.156)		(-2.119)	
R&D	0.311	***	0.440	***	0.311	***	0.409	***
	(7.262)		(8.637)		(7.987)		(7.913)	
R&D missing	-0.116	***	-0.108	***	-0.071	***	-0.061	***
	(-7.705)		(-6.654)		(-5.112)		(-4.046)	
Insider Ownership			0.135				0.131	
			(1.315)				(1.087)	
Asset Growth					-0.126	***	-0.135	***
					(-7.618)		(-7.161)	
Sales Growth					0.103	***	0.077	***
					(4.441)		(2.638)	
S&P 500					0.205	***	0.202	***
					(13.534)		(11.235)	
Leverage					-0.022		0.002	
					(-0.608)		(0.045)	
Profit Margin					0.283	***	0.304	***
					(6.978)		(6.437)	
Stock Illiquidity					-0.371	***	-1.304	***
					(-4.177)		(-9.174)	
Liquid Assets					0.143	***	0.129	**
					(3.083)		(2.494)	
Number of Deals					0.000		0.000	
					(0.885)		(0.130)	
No Deals					-0.034	**	-0.029	*
					(-2.535)		(-1.873)	
Industry Sales Share					0.497	***	0.552	***
					(4.232)		(4.282)	
Institutional Holdings							-0.183	***
							(-5.138)	
N	27,016		21,806		23,962		16,599	
Firms	2,961		2,456		2,593		2,006	
R2	0.408		0.442		0.435		0.479	

Which statement regarding the table is **not** correct?

a) In Column (1), the coefficient on the Staggered Board dummy variable is negative and statistically significant suggesting that a staggered board is associated with lower firm values.
b) Such a negative relation between a staggered board and firm value is to be expected given that a staggered board makes a takeover of the company less attractive: Ultimately, it will take two years before an acquirer will be able to obtain board majority. Hence, CEOs are to a certain extent protected against the "market for corporate control".
c) Adding more control variables results in the negative and significant relation between the staggered board dummy variable and firm value turning insignificant.
d) The additional control variables added to Columns (3) and (4) appear to absorb the previously significant association between the Staggered Board variable and firm value.
e) None of the answers a) to d) are correct.

1. Quadratic Formula

$$ax^2 + bx + c = 0 \quad \Rightarrow \quad x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Finite Series of Payments

$$PV = \sum_{t=1}^{N} \frac{C}{(1+r)^t} = \frac{C}{r} - \frac{1}{(1+r)^N} \cdot \frac{C}{r} = \left(1 - \frac{1}{(1+r)^N}\right) \cdot \frac{C}{r}$$

3. Finite Series of Payments with Growth

$$PV = \frac{C}{1+r} + \frac{C \cdot (1+g)}{(1+r)^2} + \frac{C \cdot (1+g)^2}{(1+r)^3} + \ldots + \frac{C \cdot (1+g)^{N-1}}{(1+r)^N} = \left(1 - \left(\frac{1+g}{1+r}\right)^N\right) \cdot \frac{C}{r-g}$$

4. Infinite Series of Payments

$$PV = \frac{C}{1+r} + \frac{C \cdot (1+g)}{(1+r)^2} + \frac{C \cdot (1+g)^2}{(1+r)^3} + \ldots = \sum_{t=1}^{\infty} \frac{C \cdot (1+g)^{t-1}}{(1+r)^t} = \frac{C}{r-g}$$

5. Continuous Returns

$$r^* = ln(1+r) \quad \Leftrightarrow \quad 1 + r = e^{r^*} = \exp(r^*)$$

6. Expected Value

$$\hat{\mu} = \frac{1}{N} \cdot \sum_{t=1}^{N} x_t$$

7. Variance and Covariance

Sample

$$\hat{\sigma}^2 = \frac{1}{N-1} \cdot \sum_{t=1}^{N} (x_t - \mu_x)^2$$

$$Cov[x_t, y_t] = \frac{1}{N-1} \cdot \sum_{t=1}^{N} (x_t - \mu_x)(y_t - \mu_y)$$

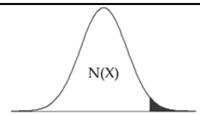
Population

$$\hat{\sigma}^2 = \frac{1}{N} \cdot \sum_{t=1}^{N} \left(x_t - E(\tilde{x})\right)^2$$

$$Cov[x_t, y_t] = \frac{1}{N} \cdot \sum_{t=1}^{N} \left(x_t - E(\tilde{x})\right)\left(y_t - E(\tilde{y})\right)$$

8. Portfolio Variance

$$Var_{Portfolio} = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j \sigma_{ij}$$



Values of the distribution function of the Standard Normal Distribution

x	N(-x)	N(x)	2·N(x) - 1
0.01	0.496	0.504	0.008
0.02	0.492	0.508	0.016
0.03	0.488	0.512	0.024
0.04	0.484	0.516	0.032
0.05	0.480	0.520	0.040
0.06	0.476	0.524	0.048
0.07	0.472	0.528	0.056
0.08	0.468	0.532	0.064
0.09	0.464	0.536	0.072
0.10	0.460	0.540	0.080
0.11	0.456	0.544	0.088
0.12	0.452	0.548	0.096
0.13	0.448	0.552	0.103
0.14	0.444	0.556	0.111
0.15	0.440	0.560	0.119
0.16	0.436	0.564	0.127
0.17	0.433	0.567	0.135
0.18	0.429	0.571	0.143
0.19	0.425	0.575	0.151
0.20	0.421	0.579	0.159
0.21	0.417	0.583	0.166
0.22	0.413	0.587	0.174
0.23	0.409	0.591	0.182
0.24	0.405	0.595	0.190
0.25	0.401	0.599	0.197
0.26	0.397	0.603	0.205
0.27	0.394	0.606	0.213
0.28	0.390	0.610	0.221
0.29	0.386	0.614	0.228
0.30	0.382	0.618	0.236
0.31	0.378	0.622	0.243
0.32	0.374	0.626	0.251
0.33	0.371	0.629	0.259
0.34	0.367	0.633	0.266
0.35	0.363	0.637	0.274
0.36	0.359	0.641	0.281
0.37	0.356	0.644	0.289
0.38	0.352	0.648	0.296
0.39	0.348	0.652	0.303
0.40	0.345	0.655	0.311
0.41	0.341	0.659	0.318
0.42	0.337	0.663	0.326
0.43	0.334	0.666	0.333
0.44	0.330	0.670	0.340
0.45	0.326	0.674	0.347
0.46	0.323	0.677	0.354
0.47	0.319	0.681	0.362
0.48	0.316	0.684	0.369
0.49	0.312	0.688	0.376
0.50	0.309	0.691	0.383

x	N(-x)	N(x)	2·N(x) - 1
0.51	0.305	0.695	0.390
0.52	0.302	0.698	0.397
0.53	0.298	0.702	0.404
0.54	0.295	0.705	0.411
0.55	0.291	0.709	0.418
0.56	0.288	0.712	0.425
0.57	0.284	0.716	0.431
0.58	0.281	0.719	0.438
0.59	0.278	0.722	0.445
0.60	0.274	0.726	0.451
0.61	0.271	0.729	0.458
0.62	0.268	0.732	0.465
0.63	0.264	0.736	0.471
0.64	0.261	0.739	0.478
0.65	0.258	0.742	0.484
0.66	0.255	0.745	0.491
0.67	0.251	0.749	0.497
0.68	0.248	0.752	0.503
0.69	0.245	0.755	0.510
0.70	0.242	0.758	0.516
0.71	0.239	0.761	0.522
0.72	0.236	0.764	0.528
0.73	0.233	0.767	0.535
0.74	0.230	0.770	0.541
0.75	0.227	0.773	0.547
0.76	0.224	0.776	0.553
0.77	0.221	0.779	0.559
0.78	0.218	0.782	0.565
0.79	0.215	0.785	0.570
0.80	0.212	0.788	0.576
0.81	0.209	0.791	0.582
0.82	0.206	0.794	0.588
0.83	0.203	0.797	0.593
0.84	0.200	0.800	0.599
0.85	0.198	0.802	0.605
0.86	0.195	0.805	0.610
0.87	0.192	0.808	0.616
0.88	0.189	0.811	0.621
0.89	0.187	0.813	0.627
0.90	0.184	0.816	0.632
0.91	0.181	0.819	0.637
0.92	0.179	0.821	0.642
0.93	0.176	0.824	0.648
0.94	0.174	0.826	0.653
0.95	0.171	0.829	0.658
0.96	0.169	0.831	0.663
0.97	0.166	0.834	0.668
0.98	0.164	0.836	0.673
0.99	0.161	0.839	0.678
1.00	0.159	0.841	0.683

x	N(-x)	N(x)	2·N(x) - 1
1.01	0.156	0.844	0.688
1.02	0.154	0.846	0.692
1.03	0.152	0.848	0.697
1.04	0.149	0.851	0.702
1.05	0.147	0.853	0.706
1.06	0.145	0.855	0.711
1.07	0.142	0.858	0.715
1.08	0.140	0.860	0.720
1.09	0.138	0.862	0.724
1.10	0.136	0.864	0.729
1.11	0.133	0.867	0.733
1.12	0.131	0.869	0.737
1.13	0.129	0.871	0.742
1.14	0.127	0.873	0.746
1.15	0.125	0.875	0.750
1.16	0.123	0.877	0.754
1.17	0.121	0.879	0.758
1.18	0.119	0.881	0.762
1.19	0.117	0.883	0.766
1.20	0.115	0.885	0.770
1.21	0.113	0.887	0.774
1.22	0.111	0.889	0.778
1.23	0.109	0.891	0.781
1.24	0.107	0.893	0.785
1.25	0.106	0.894	0.789
1.26	0.104	0.896	0.792
1.27	0.102	0.898	0.796
1.28	0.100	0.900	0.799
1.29	0.099	0.901	0.803
1.30	0.097	0.903	0.806
1.31	0.095	0.905	0.810
1.32	0.093	0.907	0.813
1.33	0.092	0.908	0.816
1.34	0.090	0.910	0.820
1.35	0.089	0.911	0.823
1.36	0.087	0.913	0.826
1.37	0.085	0.915	0.829
1.38	0.084	0.916	0.832
1.39	0.082	0.918	0.835
1.40	0.081	0.919	0.838
1.41	0.079	0.921	0.841
1.42	0.078	0.922	0.844
1.43	0.076	0.924	0.847
1.44	0.075	0.925	0.850
1.45	0.074	0.926	0.853
1.46	0.072	0.928	0.856
1.47	0.071	0.929	0.858
1.48	0.069	0.931	0.861
1.49	0.068	0.932	0.864
1.50	0.067	0.933	0.866

x	N(-x)	N(x)	2·N(x) - 1
1.51	0.066	0.934	0.869
1.52	0.064	0.936	0.871
1.53	0.063	0.937	0.874
1.54	0.062	0.938	0.876
1.55	0.061	0.939	0.879
1.56	0.059	0.941	0.881
1.57	0.058	0.942	0.884
1.58	0.057	0.943	0.886
1.59	0.056	0.944	0.888
1.60	0.055	0.945	0.890
1.61	0.054	0.946	0.893
1.62	0.053	0.947	0.895
1.63	0.052	0.948	0.897
1.64	0.051	0.949	0.899
1.65	0.049	0.951	0.901
1.66	0.048	0.952	0.903
1.67	0.047	0.953	0.905
1.68	0.046	0.954	0.907
1.69	0.046	0.954	0.909
1.70	0.045	0.955	0.911
1.71	0.044	0.956	0.913
1.72	0.043	0.957	0.915
1.73	0.042	0.958	0.916
1.74	0.041	0.959	0.918
1.75	0.040	0.960	0.920
1.76	0.039	0.961	0.922
1.77	0.038	0.962	0.923
1.78	0.038	0.962	0.925
1.79	0.037	0.963	0.927
1.80	0.036	0.964	0.928
1.81	0.035	0.965	0.930
1.82	0.034	0.966	0.931
1.83	0.034	0.966	0.933
1.84	0.033	0.967	0.934
1.85	0.032	0.968	0.936
1.86	0.031	0.969	0.937
1.87	0.031	0.969	0.939
1.88	0.030	0.970	0.940
1.89	0.029	0.971	0.941
1.90	0.029	0.971	0.943
1.91	0.028	0.972	0.944
1.92	0.027	0.973	0.945
1.93	0.027	0.973	0.946
1.94	0.026	0.974	0.948
1.95	0.026	0.974	0.949
1.96	0.025	0.975	0.950
1.97	0.024	0.976	0.951
1.98	0.024	0.976	0.952
1.99	0.023	0.977	0.953
2.00	0.023	0.977	0.954

x	N(-x)	N(x)	2·N(x) - 1
2.01	0.022	0.978	0.956
2.02	0.022	0.978	0.957
2.03	0.021	0.979	0.958
2.04	0.021	0.979	0.959
2.05	0.020	0.980	0.960
2.06	0.020	0.980	0.961
2.07	0.019	0.981	0.962
2.08	0.019	0.981	0.962
2.09	0.018	0.982	0.963
2.10	0.018	0.982	0.964
2.11	0.017	0.983	0.965
2.12	0.017	0.983	0.966
2.13	0.017	0.983	0.967
2.14	0.016	0.984	0.968
2.15	0.016	0.984	0.968
2.16	0.015	0.985	0.969
2.17	0.015	0.985	0.970
2.18	0.015	0.985	0.971
2.19	0.014	0.986	0.971
2.20	0.014	0.986	0.972
2.21	0.014	0.986	0.973
2.22	0.013	0.987	0.974
2.23	0.013	0.987	0.974
2.24	0.013	0.987	0.975
2.25	0.012	0.988	0.976
2.26	0.012	0.988	0.976
2.27	0.012	0.988	0.977
2.28	0.011	0.989	0.977
2.29	0.011	0.989	0.978
2.30	0.011	0.989	0.979
2.31	0.010	0.990	0.979
2.32	0.010	0.990	0.980
2.33	0.010	0.990	0.980
2.34	0.010	0.990	0.981
2.35	0.009	0.991	0.981
2.36	0.009	0.991	0.982
2.37	0.009	0.991	0.982
2.38	0.009	0.991	0.983
2.39	0.008	0.992	0.983
2.40	0.008	0.992	0.984
2.41	0.008	0.992	0.984
2.42	0.008	0.992	0.984
2.43	0.008	0.992	0.985
2.44	0.007	0.993	0.985
2.45	0.007	0.993	0.986
2.46	0.007	0.993	0.986
2.47	0.007	0.993	0.986
2.48	0.007	0.993	0.987
2.49	0.006	0.994	0.987
2.50	0.006	0.994	0.988

x	N(-x)	N(x)	2·N(x) - 1
2.51	0.006	0.994	0.988
2.52	0.006	0.994	0.988
2.53	0.006	0.994	0.989
2.54	0.006	0.994	0.989
2.55	0.005	0.995	0.989
2.56	0.005	0.995	0.990
2.57	0.005	0.995	0.990
2.58	0.005	0.995	0.990
2.59	0.005	0.995	0.990
2.60	0.005	0.995	0.991
2.61	0.005	0.995	0.991
2.62	0.004	0.996	0.991
2.63	0.004	0.996	0.991
2.64	0.004	0.996	0.992
2.65	0.004	0.996	0.992
2.66	0.004	0.996	0.992
2.67	0.004	0.996	0.992
2.68	0.004	0.996	0.993
2.69	0.004	0.996	0.993
2.70	0.003	0.997	0.993
2.71	0.003	0.997	0.993
2.72	0.003	0.997	0.993
2.73	0.003	0.997	0.994
2.74	0.003	0.997	0.994
2.75	0.003	0.997	0.994
2.76	0.003	0.997	0.994
2.77	0.003	0.997	0.994
2.78	0.003	0.997	0.995
2.79	0.003	0.997	0.995
2.80	0.003	0.997	0.995
2.81	0.002	0.998	0.995
2.82	0.002	0.998	0.995
2.83	0.002	0.998	0.995
2.84	0.002	0.998	0.995
2.85	0.002	0.998	0.996
2.86	0.002	0.998	0.996
2.87	0.002	0.998	0.996
2.88	0.002	0.998	0.996
2.89	0.002	0.998	0.996
2.90	0.002	0.998	0.996
2.91	0.002	0.998	0.996
2.92	0.002	0.998	0.996
2.93	0.002	0.998	0.997
2.94	0.002	0.998	0.997
2.95	0.002	0.998	0.997
2.96	0.002	0.998	0.997
2.97	0.001	0.999	0.997
2.98	0.001	0.999	0.997
2.99	0.001	0.999	0.997
3.00	0.001	0.999	0.997